

INTERNATIONAL GCSE **CHEMISTRY**

9202/1

Paper 1

Mark scheme

June 2025

Version: 0.1 Pre-Standardisation



Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available ‘any **two** from’ is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that ‘right + wrong = wrong’.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols / formulae

If a student writes a chemical symbol / formula instead of a required chemical name, full credit can be given if the symbol / formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(...) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

'Ignore' is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do not accept

'Do **not** accept' means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array} $		1	AO1 3.10.3.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	fermentation		1	AO1 3.10.1.2f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	7		1	AO1 3.10.3.1b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	oxidation		1	AO1 3.10.3.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	3		1	AO2 3.10.3.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.6	C ₆ H ₈ O ₇		1	AO2 3.6.2a 3.10.3.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.7	(a weak acid) does not ionise completely (when dissolved) in water or (a weak acid) does not ionise completely in aqueous solution	allow (a weak acid) partially dissociates (when dissolved) in water allow (a weak acid) partially dissociates in aqueous solution	1	AO1 3.10.3.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.8	any one from: • so the medicine is safe • impurities may be dangerous / toxic • to avoid side-effects (from impurities)	 allow impurities can damage the body	1	AO3 3.4.1b

Total Question 1		8
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Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	(the symbol) $\rightleftharpoons$	allow description of reversible sign	1	AO1 3.9.1d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	white		1	AO1 3.9.1d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	add water		1	AO1 3.9.1d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	from 1.31 (g) to 1.39 (g)	allow from 1.39 (g) to 1.31 (g)	1	AO4 3.9.1d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	water / steam escapes		1	AO3 3.6.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.6	reheat (and reweigh) until constant mass	 if no other mark awarded allow for 1 mark heat for longer or (heat at a) higher temperature	1 1	AO4 3.6.1e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.7	(test) (add) barium chloride (solution)	allow (add) barium nitrate (solution)	1	AO1 3.4.3f
	(in the presence of) hydrochloric acid	allow (in the presence of) nitric acid	1	
	(result) white precipitate produced	MP2 is dependent on MP1 being awarded MP3 is dependent on MP1 being awarded	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.8	(mass of water in 3.90 g = 3.90 – 2.50 =) 1.40 (g)		1	AO2 3.6.1c 3.9.1d
	(mass of sample =) $15.12 \times \frac{3.90}{1.40}$	allow correct use of an incorrectly determined mass of water in 3.90 g	1	
	= 42.1 (g)	allow 42.12 (g)	1	

Total Question 2		13
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Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	1–100 nm		1	AO1 3.2.4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	a few hundred		1	AO1 3.2.4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3	(volume =) $3 \times 3 \times 3$	allow correct use of an incorrectly determined volume	1	AO2 3.2.4
	= 27 (nm ³)		1	
	(surface area : volume =) 54 : 27		1	
	(simplest ratio) = 2 : 1		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	(nanoparticles) have a higher surface area to volume ratio or less (material) is needed (for the same effect)	allow converse for larger particles	1	AO3 3.2.4

Total Question 3		7
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Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	about 100		1	AO1 3.1.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	atomic number		1	AO1 3.1.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	any three from: (Mendeleev's Group 1) • contains copper / Cu • contains silver / Ag • contains hydrogen / H • has some boxes with two elements • does not show atomic number • does not show mass number • does not show caesium / Cs • does not show francium / Fr	allow converse	3	AO3 3.1.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.4	(element) gallium / Ga (reason) (gallium) has a low melting point (gallium) has a similar density (gallium) has the same formula of oxide	allow 5.9 is closest to 6.0 if no other marks awarded allow 1 mark for scandium has the same formula of oxide	1 1 1 1	AO3 3.1.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.5	(relative formula mass of 2D = $484 - (16 \times 7) =$ 372 (relative atomic mass of D = $\frac{372}{2} =$) 186 (element D) rhenium / Re	allow correct use of incorrectly determined value of 2D allow correctly determined element from an incorrectly determined relative atomic mass	1 1 1	AO2 3.6.2a

Total Question 4		12
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Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	4		1	AO1 3.2.3a

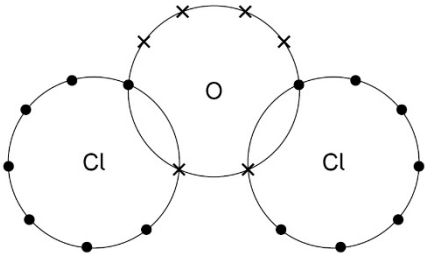
Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	very hard		1	AO1 3.2.3b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	4000 °C		1	AO2 3.2.2f

Question	Answers	Mark	AO/ Spec. Ref.
05.4	Level 3: Relevant points (reasons / causes) are identified, given in detail and logically linked to form a clear account.	5–6	AO1 3.2.3c,d
	Level 2: Relevant points (reasons / causes) are identified, and there are attempts at logical linking. The resulting account is not fully clear.	3–4	
	Level 1: Points are identified and stated simply, but their relevance is not clear and there is no attempt at logical linking.	1–2	
	No relevant content	0	
	Indicative content: Structure and bonding - giant structure / lattice (of carbon atoms) - covalent (bonds) - strong (covalent) bonds - in layers - of hexagonal rings Explanation for being slippery - layers free to slide over each other (because) no covalent bonds between layers or (because) only weak (intermolecular) forces between layers Explanation for conductivity - each (carbon) atom bonded to three other (carbon) atoms - one electron from each atom is delocalised - delocalised electrons (which) move through the structure (which) carry charge		

Total Question 5		9
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Question 6

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.1	1 pair of shared electrons in each overlap 6 unbonded electrons on each chlorine and 4 unbonded electrons on oxygen	allow any combination of x, •, o, or e⁽⁻⁾ for electrons an answer of  scores 2 marks	1 1	AO2 3.2.1a,b,g

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	magnesium / chlorine would react with the oxygen (in the air)		1	AO2 3.2.1a,c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	magnesium atom loses two electrons	} allow 1 mark for magnesium loses electrons and chlorine gains electron(s)	1	AO2 3.2.1b,c
	chlorine atoms gain one electron		1	
	two chlorine atoms react with one magnesium atom		1	
	to form ions that have the electron arrangement of a noble gas	allow to gain full outer shells of electrons allow to form Mg^{2+} ions and (two) Cl^- ions	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.4	(test) add sodium hydroxide solution	MP2 is dependent upon MP1 being awarded	1	AO1 3.4.3b
	(result) white precipitate		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.5	selenium / Se		1	AO2 3.1.2i 3.2.1c

Question	Answers				Extra information	Mark	AO/ Spec. Ref.
06.6		Mg	O	S	allow correct use of an incorrectly determined number of moles		AO2 3.6.2c
	(moles =)	$\frac{8.40}{24}$	$\frac{16.80}{16}$	$\frac{22.40}{32}$		1	
		= 0.35	= 1.05	= 0.70		1	
		$\frac{0.35}{0.35} = 1$	$\frac{1.05}{0.35} = 3$	$\frac{0.70}{0.35} = 2$		1	
	(empirical formula) MgO_3S_2				allow correct use of an incorrectly determined ratio of moles	1	

Total Question 6		14
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Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	(nitrogen) air		1	AO1 3.8.3a
	(hydrogen) natural gas		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	iron		1	AO1 3.8.3c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	(temperature) a value in the range –196 to –33 (°C)		1	AO3 3.8.3c
	(reason) at this temperature only ammonia condenses		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.4	recycled	allow reused	1	AO1 3.8.3c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.5	no reactants or products can enter or leave the system		1	AO2 3.8.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.6	the reactions occur at exactly the same rate in each direction	allow the concentrations of the products and reactants are constant	1	AO1 3.8.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.7	(as temperature increases) the percentage of ammonia at equilibrium decreases		1	AO2 × 2 AO3 × 1 3.8.2c,e
	(because the forward) reaction is exothermic		1	
	(so the) equilibrium position shifts to left hand side		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.8	(as pressure increases) the percentage of ammonia at equilibrium increases		1	AO2 × 2 AO3 × 1 3.8.2d,e
	(because) an increase in pressure favours the forward reaction	allow (because) the equilibrium shifts to the right-hand side	1	
	(because) fewer moles on the right-hand side or (because) the forward reaction produces the fewest number of molecules		1	

Total Question 7		14
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Question 8

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.1	the ions are free to move		1	AO1 3.3.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.2	any one from: • appearance • durability • prevention of corrosion		1	AO1 3.3.2h

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.3	$2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$	allow multiples allow for 1 mark $\text{Cl}^- \rightarrow \text{Cl}_2 + \text{e}^-$	2	AO1 3.3.2f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.4	14	allow a value in the range 8 to 14	1	AO1 3.5.1a,f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.5	soap		1	AO1 3.3.2j

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.6	(conversion $600 \text{ cm}^3 =$) $0.6 \text{ (dm}^3\text{)}$		1	AO2 3.6.3a 3.6.4a
	(moles of NaOH =) $0.35 \times 0.6 = 0.21$	allow correct use of incorrect / no conversion of volume	1	
	(mass of NaOH =) 0.21×40	allow correct use of incorrectly determined value for moles	1	
	= 8.4 (g)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.7	hydrogen is less reactive than sodium		1	AO2 3.3.2g,j

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.8	hydrogen ions gain electrons to form a hydrogen molecule	allow hydrogen ions are reduced	1	AO2 3.3.2e,j
			1	

Total Question 8		13
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